

UTS-UAV: A Context-Grounded Benchmark for Operational Event Discovery in Low-Altitude UAV Imagery

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Abstract—Operational UAV inspection must reject ordinary frames, locate contextual evidence, and recognize an event; classifying a supplied crop tests only part of this decision. We introduce UTS-UAV, a real low-altitude benchmark linking full frames, human-accepted contextual ROIs, high-resolution crops, event labels, and genuine no-event negatives. Five tasks separate presence-aware grounding, two oracle recognition settings, joint discovery, and modular discovery. Grouped protocols define session-disjoint, unseen-site, and forward-temporal evaluation. In the completed session-disjoint study of 4,778 positive images, one shared Qwen3-VL classifier reaches 72.24% macro-F1 on GT crops and 71.98% with ROI plus context, but only 67.34% with coordinates. We introduce Qwen-Ground, a visual-only baseline that predicts presence and a context box through a multi-scale Transformer decoder, then combines feature-space ROI and global evidence for recognition without language decoding. Eight full-frame systems are evaluated on the same 4,778 positive and 560 no-event records. Direct generative Qwen attains 6.10% event-aware mAP50 but produces 46.61% no-event false positives at 99.79% positive recall. Qwen-Ground reaches 3.57% mAP50 with 15.89% false positives and 79.87% recall; adaptive tiling reaches 5.52% mAP50 with 3.75% false positives but only 57.68% recall. These complementary operating points expose the gap between oracle recognition and reliable full-frame discovery, and show why localization quality and the false-alarm/recall tradeoff must be evaluated together.

I. INTRODUCTION

Low-altitude UAVs can survey farmland, waterways, construction zones, and settlements at a scale that is impractical for ground patrols. A deployable inspection system must answer three coupled questions: *is* an event present, *where* is the evidence-bearing region, and *what* event does it represent? The first question is easily hidden by positive-only datasets; the second is difficult because useful evidence may occupy a tiny fraction of a 4K frame; and the third is contextual because labels such as exposed waste, unclean water, or active construction describe conditions, not pixel-tight object categories.

Existing aerial datasets are rich but leave this combination under-specified. DOTA and VisDrone emphasize generic objects under scale and viewpoint variation [1], [2]; ERA and CapERA study event recognition or description from selected clips [3], [4]. Recent UAV vision–language benchmarks greatly broaden semantic, spatial, and planning evaluation [5]–[7], while AirZoo contributes geometry-rich supervision and cross-domain transfer tracks [8]. These advances do not directly quantify the failure chain from ordinary-scene rejection to context-region discovery and fine-grained operational recognition. A crop classifier can therefore appear deployment-ready without ever searching a full frame or facing a true negative.

The closest recent grounding benchmarks sharpen this gap. AerialVG and DVGBench test explicit or implicit references in high-resolution drone imagery [9], [10]; UniRef-UAV additionally admits absent targets and variable output cardinality [11]. Their targets remain referred objects. In operational inspection, by contrast, the query is implicit (“find a reportable event”), the positive region may be a diffuse condition, and the first correct output for many frames is no-event.

We build UTS-UAV from a production inspection archive while treating its workflow provenance as part of the benchmark contract. The source platform connects mission dispatch and multi-source image ingestion to AI-assisted retrieval, operator review, alerting, replay, and reporting. We distill that stream rather than its proprietary interface: full-frame/crop pairs are linked, contextual boxes are recovered and manually accepted, operational labels are frozen, and genuine no-event frames are retained. Content, session, and site groups are assigned before splitting, and test labels are sealed. The benchmark consequently supports controlled counterfactuals: a crop estimates oracle recognition performance; a ground-truth ROI tests whether full-frame compression and background are harmful; presence and class-agnostic grounding measure search; and joint or cascaded inference measures the perception front end of an alert–review–report loop.

Following the property–track organization of large dataset papers and the capability–diagnostic structure of recent UAV benchmarks, we make four contributions:

- **A context-grounded operational benchmark.** UTS-UAV combines true positive and no-event frames with full-resolution imagery, contextual ROIs, crops, event labels, and leakage-safe acquisition groups.
- **Five interoperable tasks and grouped protocols.** A single record contract supports presence-aware grounding, two oracle recognition settings, unified discovery, and cascaded discovery. Session, site, and temporal protocols separate evaluation settings; completed results concern sessions.
- **A controlled representation study.** On the same 4,778 session-disjoint positive records, we compare crop, contextual, drawn-box, coordinate, and masked views with one shared Qwen adaptation, alongside ConvNeXt, DINOv2, and Florence-2 references.
- **A full-frame feature-space baseline and diagnostic suite.** Qwen-Ground couples Qwen visual tokens to multi-scale contextual grounding, explicit presence prediction, feature-space ROI pooling, and definition-prototype classification. Eight completed full-frame

One record. Two resolutions. Contextual event evidence.

Real full-frame / crop pairs; boxes show accepted contextual ROIs, not object boundaries.

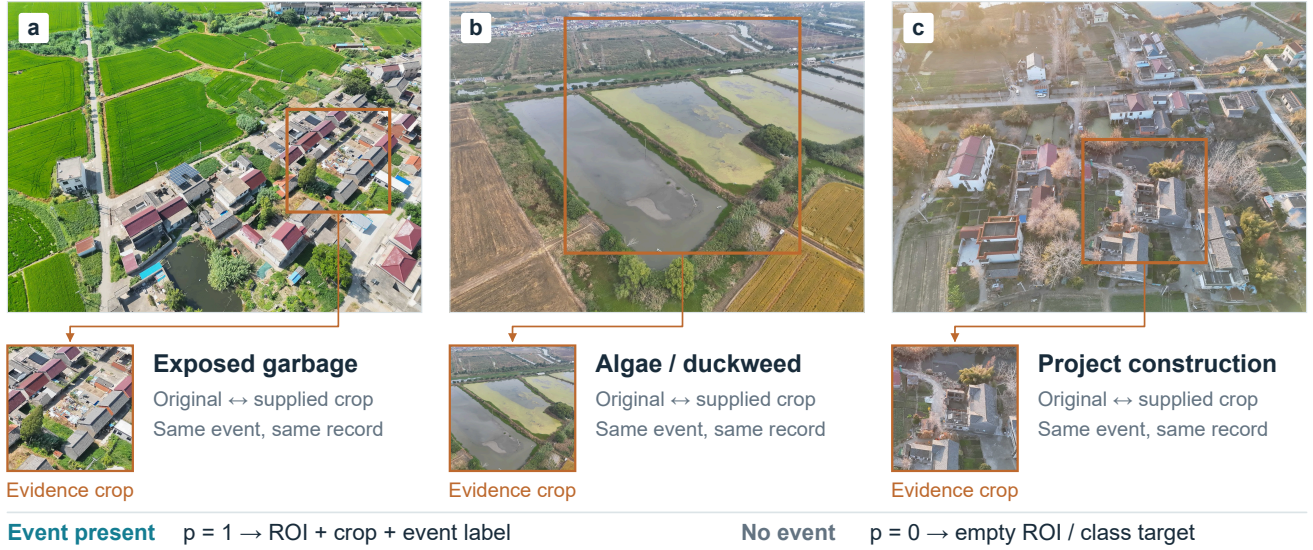


Fig. 1. UTS-UAV evaluates full-frame operational event discovery. Positive examples (shown) link a full frame, a human-accepted contextual ROI, and a supplied crop: exposed garbage, algae/duckweed, and project construction. Boxes denote sufficient contextual evidence rather than pixel-tight objects. The bottom strip states the positive/no-event target contract; it does not depict a negative image or a model prediction.

systems expose the tradeoff between false alarms, recall, and event-aware grounding.

II. RELATED WORK

Aerial perception and operational events. DOTA and VisDrone established aerial object detection under scale, orientation, and density variation [1], [2]. ERA and CapERA extend the task to aerial event recognition and captioning [3], [4], while A2Seek studies video anomaly reasoning [12]. UTS-UAV instead targets static operational inspection, linking ordinary-scene rejection to contextual-region discovery and recognition. Its ROIs may include terrain, water, or neighboring structures because a reportable condition is not necessarily a single physical object.

UAV foundation-model benchmarks. UAVBench, MM-UAVBench, and SpatialUAV broaden evaluation across image/region understanding, planning, and spatial intelligence [5]–[7]. AirZoo organizes aerial retrieval, cross-view matching, and reconstruction around large-scale geometric pretraining and real-domain transfer [8]. These systematic task and transfer taxonomies motivate our five linked tasks, but our evaluation follows a different decision chain: no-event, context ROI, and operational class. Table I summarizes the scope distinction.

Grounding and unified visual prediction. Grounding DINO and YOLO-World provide open-vocabulary object localization [13], [14]; D-FINE is a closed-set detection reference [15]. Florence-2 and Qwen3-VL support prompt-driven visual predictions [16], [17], while DINOv2 supplies transferable visual features [18]. AerialVG studies relational grounding in high-resolution drone imagery, DVGBench targets implicit references, and UniRef-UAV includes absent

targets [9]–[11]. These are important neighbors, but referred objects differ from diffuse, implicitly queried operational events. We therefore evaluate both class-agnostic contextual search and event-aware discovery, retaining negative frames rather than presuming a target is present.

Correlated acquisition and false alarms. Repeated captures can make random image splits reward place or trajectory memorization. Our grouped protocols separate acquisition content and operational sessions, with site/time protocols defining further tests. Aerial Mirage documents aerial LVLMM hallucination [19], reinforcing the need to score unsupported event predictions on ordinary frames. We pair positive recall with negative-frame false-positive rate instead of equating high positive-only recognition with reliable inspection.

III. UTS-UAV BENCHMARK

A. Operational System Provenance

The source archive was exported, under company authorization, from a production inspection platform rather than assembled by web search. Scheduled and ad-hoc missions feed extracted video frames, image libraries, and local uploads into an AI-assisted alert–review–report loop. This creates correlated captures, ordinary frames that never become alerts, and positives whose meaning depends on task and scene context. The operational taxonomy even separates similar material by setting—for example land, water, collection-point, and bank-side garbage, or construction- versus factory-side storage. Such distinctions motivate contextual ROIs: a tight debris or material box can erase the evidence that defines the reportable condition.

TABLE I

POSITIONING AGAINST REPRESENTATIVE AERIAL BENCHMARKS. “NO-EVENT” MEANS EXPLICIT ORDINARY/NEGATIVE FRAMES ARE PART OF THE TARGET EVALUATION; CONTEXT ROI DENOTES A SUFFICIENT-EVIDENCE REGION RATHER THAN A PIXEL-TIGHT OBJECT BOX. SHIFT REQUIRES GROUPED GEOGRAPHIC OR TEMPORAL TESTING.

Benchmark	Main scope	Real UAV	Region output	No-event	Context ROI	Full→event discovery	Site/time shift
ERA [3]	event recognition	✓	–	–	–	–	–
VisDrone [2]	object detection/tracking	✓	✓	–	–	–	–
A2Seek [12]	video anomaly reasoning	✓	✓	✓	–	✓	–
UAVBench [5]	image/region VL	✓	✓	–	–	–	–
AerialVG [9]	relational grounding	✓	✓	–	–	–	–
UniRef-UAV [11]	universal referring	✓	✓	✓	–	–	partial
MM-UAVBench [6]	perception–planning	✓	partial	–	–	–	–
SpatialUAV [7]	spatial intelligence	✓	✓	–	–	–	–
UTS-UAV	operational events	✓	✓	✓	✓	✓	✓

B. Three Benchmark Properties

Operational realism. UTS-UAV contains **TBD** full-resolution frames, comprising **TBD** event-bearing samples, **TBD** true no-event samples, and 18 operational event classes. Negatives are captured by the same operational process rather than synthesized from random background patches, enabling meaningful false-alarm and calibration analysis.

Context-grounded supervision. A sample is

$$d_i = (I_i, p_i, b_i, C_i, y_i, m_i), \quad (1)$$

where I_i is the full frame, $p_i \in \{0, 1\}$ is event presence, b_i is a contextual evidence ROI, C_i is its high-resolution crop, y_i is the event label, and m_i contains opaque acquisition groups. For $p_i = 0$, $(b_i, C_i, y_i) = \emptyset$; for $p_i = 1$, the full/ROI/crop views share one record identity. The ROI encloses evidence sufficient for an operator to judge the event. It can include surrounding water, terrain, buildings, or activity context and is not a pixel-tight physical-object boundary.

Deployment-oriented protocols. Content, acquisition session, site, and time groups are frozen before model development. This supports controlled evaluation on repeated operations, unseen geography, and future capture periods without allowing near-duplicate frames to cross a boundary.

C. Construction and Quality Assurance

Positive full/crop pairs are linked using delivery keys and media hashes. Crop-to-frame registration proposes each ROI using local features, geometric verification, and corner projection; every proposed box is subsequently accepted or corrected by human review. Negatives are verified as containing no benchmark event under the frozen ontology. Label adjudication retains the source operational distinctions, with a fixed 18-class ontology. Company authorization, privacy review, release redaction, and test-label sealing are completed. Public records expose only opaque session/site identifiers and the metadata necessary to reproduce the protocols.

D. Five Controlled Tasks

T1: presence-aware contextual grounding maps a full frame to either no-event or one ROI. **T2: oracle-box recognition** supplies a positive full frame and its GT ROI as

From operations to a benchmark

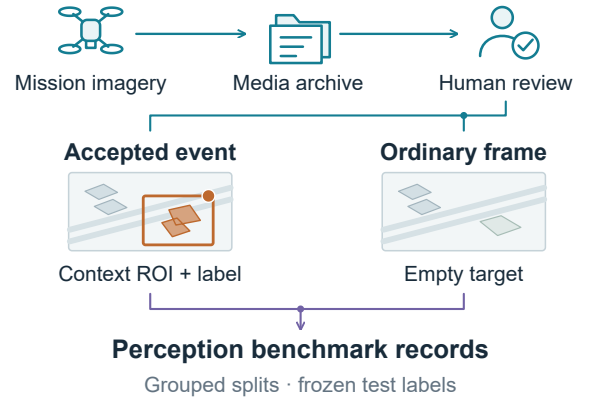


Fig. 2. Source-to-benchmark abstraction. Reviewed events and ordinary frames produce contextual-ROI and empty targets, respectively. Both enter grouped perception evaluation; dispatch and remediation are outside its scope. Scene thumbnails are schematic, not dataset examples or proprietary screens.

TABLE II

FIVE LINKED TASKS. POS.: POSITIVE EVENT FRAME; NEG.: NO-EVENT FRAME.

Task	Input	Output	Pos.	Neg.
T1	full frame	no-event or ROI	✓	✓
T2	full + GT ROI	event class	✓	–
T3	GT crop	event class	✓	–
T4	full frame	no-event or ROI+event	✓	✓
T5	full→pred. crop	no-event or ROI+event	✓	✓

drawn, coordinate, or masked-background prompt. **T3: oracle-crop recognition** classifies the delivered positive crop and supplies an oracle reference. **T4: unified discovery** returns no-event or an ROI–event pair in one call. **T5: modular discovery** first proposes no-event/ROI and then classifies a source-resolution crop. T2–T3 isolate representation and context; T1 isolates presence/search; T4–T5 compare unified and modular deployment while sharing the same records.

E. Leakage-Controlled Protocols

The primary **session-disjoint** protocol assigns entire acquisition and content groups to train, validation, or test (**TBD/TBD/TBD** frames). **Unseen-site** holds opaque geographic groups out for test (**TBD/TBD/TBD**), and **forward-temporal** trains on past captures and tests on later months (**TBD/TBD/TBD**). Positive records obey the corresponding content/session/site/time constraint. The 2,534 no-event frames use one fixed group-disjoint partition (1,630/344/560, with 3/1/3 source groups) shared by all three protocols. Thus negative evaluation is group-disjoint, but we do not attribute site or temporal semantics to negative retention until aligned metadata are available. Every derived task keeps the parent record assignment, and all model, threshold, prompt, and context-margin selection is validation-only.

F. Metrics and Evaluation Contract

Each system supplies a presence score and at most one candidate context box, with one predicted class for T4/T5. Boxes use a common 0–1000 coordinate system. The evaluator retains every test record; absent boxes cannot produce localization matches. Current detector results use their saved native coordinates, including out-of-frame extents, without retrospective clipping.

Ranking. P-AP ranks all records by presence score. AP50 uses the same score to rank candidate boxes, requiring a positive target and $\text{IoU} \geq 0.5$. G-mAP50 additionally requires the correct event class: only the single predicted class receives that candidate, and AP is macro-averaged over 18 classes. Candidates on negatives count as false positives. For each ranked list, non-interpolated AP is

$$\text{AP} = \frac{1}{N_{\text{GT}}} \sum_{k=1}^M \text{Prec}(k) \mathbf{1}[\text{match}_k], \quad (2)$$

where N_{GT} counts all eligible ground-truth positives, including those with missing predictions. These are single-candidate contextual AP metrics, not COCO multi-instance mAP. Ranked metrics are computed before presence thresholding.

Operating point. A frame is accepted when its presence score is at least its saved threshold. We report both $\text{N-FPR} = \text{FP}/N_-$ and positive recall TP/N_+ . The calibration rule maximizes validation recall subject to $\text{N-FPR} \leq 10\%$, breaking ties by higher precision and then higher threshold. Thresholds are frozen before test; the legacy YOLO-World zero-threshold exception is explicitly marked in Table IV. The direct-Qwen and tiling calibration cache contains 2,384 positives and 344 negatives, with no test-ID overlap. The saved Qwen-Ground calibration configuration uses a 512-positive subset and the 344 negatives; this difference prevents a strictly matched calibration-budget claim.

Recognition and uncertainty. C-F1 is macro-F1 across the 18 candidate classes on positive frames, before presence gating; missing classes remain errors. T2/T3 use the same class ontology but oracle regions. All current results are seed-43 point estimates. Neither group-bootstrap intervals nor training-seed variation is claimed without corresponding completed analyses.

IV. FEATURE-SPACE QWEN-GROUND

Qwen-Ground uses the visual tower from the Qwen3.6-35B-A3B-FP8 checkpoint; direct generative baselines use Qwen3-VL-8B [17]. It separates three questions that generative coordinate strings entangle: whether an event exists, where its sufficient context lies, and which event definition best matches the grounded evidence. The registered input cap is 995,328 pixels. The language decoder is never called.

A. Multi-Scale Contextual Grounding

Let F_q denote spatial tokens after Qwen’s native merger. A learned projection maps them to 768 dimensions. Adaptive average pooling forms a three-level pyramid at scales $s \in \{1, 2, 4\}$; two-dimensional coordinate and level embeddings retain location and scale:

$$F_s = \text{AAP}_s(W_q F_q) + E_{xy,s}. \quad (3)$$

A single event query cross-attends to the concatenated pyramid through a one-layer, 12-head Transformer decoder [20]. On the finest grid, cosine similarity between the decoded event feature z_e and token f_i defines a spatial distribution a_i . The box center is a soft-argmax; width and height use the event feature and heatmap-weighted visual evidence:

$$\begin{aligned} a_i &= \text{softmax}_i(\tau \cos(z_e, f_i)), \\ \hat{c} &= \sum_i a_i p_i, \\ (\hat{w}, \hat{h}) &= \sigma \left(g_b[z_e, \sum_i a_i f_i] \right). \end{aligned} \quad (4)$$

A separate linear head gives event presence $\hat{p} = \sigma(w_p^\top z_e)$. This is one event query plus a presence head, not a null/event query bank.

B. Feature-Space ROI Recognition

The predicted box indexes the existing finest feature map through 4×4 ROIAlign. Spatial averaging yields a local feature z_r without re-encoding source pixels. A two-layer MLP combines z_r with the global event feature. Its normalized projection is compared by cosine similarity with 18 fixed class-definition prototypes t_k :

$$\begin{aligned} z_r &= \text{Avg}(\text{ROIAlign}(F_1, \hat{b}, 4, 4)), \\ \ell_k &= \gamma \cos(W_{\text{cg}}[z_e, z_r], t_k). \end{aligned} \quad (5)$$

Removing either z_r or z_e is therefore an interpretable ablation of local or global evidence. A fixed linear classifier replaces the prototype head in a separate control; its fixed vocabulary does not freeze its learned weights.

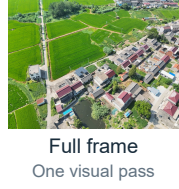
C. Optimization and Inference

Localization uses $5\mathcal{L}_1 + 2\mathcal{L}_{\text{IoU}}$ and a spatial-KL loss on the heatmap; presence uses focal loss and positive records additionally use cross-entropy classification:

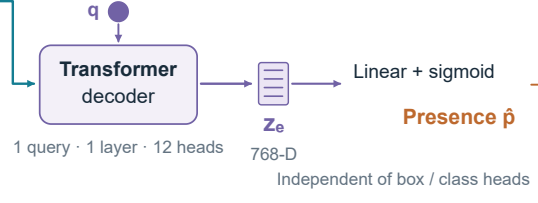
$$\mathcal{L} = \mathcal{L}_{\text{focal}} + p(5\mathcal{L}_1 + 2\mathcal{L}_{\text{IoU}} + \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{CE}}). \quad (6)$$

Training first establishes multi-scale grounding and then attaches the recognition branch. The final three vision blocks

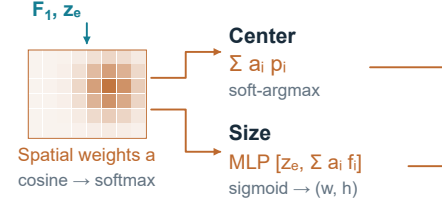
A Shared visual evidence



B Event query

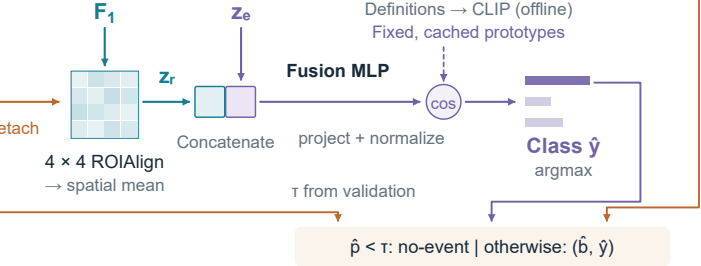


C Contextual grounding



Repeated F_1 / z_e labels denote the same shared tensors.
Feature grids, weights and class bars are schematic.

D Recognition on shared features



Training: MS 12 $\rightarrow$ CLS 12 epochs; ROI positives use GT $\rightarrow$ detached predicted boxes. No image recrop or language decoding.

Fig. 3. Qwen-Ground inference. A single visual pass supplies multi-scale query decoding, contextual grounding, and feature-space recognition. Repeated F_1/z_e labels are shared-tensor ports: ROIAlign receives feature values and detached box coordinates separately, while classification combines local and global evidence with fixed CLIP definition prototypes. Presence gates the final record, not branch execution. Grids and class bars are schematic; no pixel recrop or language decoder is used.

are updated; earlier vision blocks stay frozen. Registered learning rates are 10^{-6} for the trainable vision blocks and 10^{-4} for new heads. Positive sampling is class balanced, and batches expose 20% no-event records. Box-conditioned classification uses GT boxes in epochs 1–3, predicted detached boxes with probabilities 1/3 and 2/3 in epochs 4 and 5, and predicted detached boxes thereafter. The current matched protocol trains 12 MS epochs followed by 12 CLS epochs with seed 43 and evaluates the saved final checkpoint; it does not claim early stopping or multi-seed uncertainty.

At inference, the validation-frozen presence threshold returns either no-event or one canonical $(\hat{b}, \hat{y})$ record. T1 exposes presence and box outputs; T4 adds the feature-space classifier. T2/T3 remain oracle-representation tasks, and T5 is an exposed modular cascade rather than another head of the same model.

V. EXPERIMENTS

We report completed session-disjoint seed-43 results, not multi-seed means. Table IV is recomputed from eight saved prediction files with the same 5,338 unique record IDs, labels, and targets, using the completed matched full control for Qwen-Ground. Pending evaluations are identified separately and do not support the conclusions below.

A. Models, Training, and Reproducibility

The completed comparison covers oracle ConvNeXt/DINOv2 classifiers, Florence-2 and Qwen3-VL; D-FINE contextual localization; open-vocabulary Grounding DINO and YOLO-World; generative discovery; a D-FINE–DINOv2 cascade; adaptive Qwen tiling; and Qwen-Ground. References accompany each model in the result tables.

Oracle-view recognition and full-frame discovery use distinct adaptations and must not be interpreted as the same checkpoint with different inference prompts.

For Table III, Qwen3-VL-8B-Instruct uses LoRA rank 16, $\alpha = 32$, dropout 0.05, five epochs, batch size 4, gradient accumulation 2, and a 995,328-pixel cap. A mixed-view sampler trains all five Qwen views; validation selects the epoch maximizing their mean macro-F1 (epoch 4). A learned classification head reads the last non-padding LLM hidden state, rather than generated text; its 100% output validity is structural. Direct T4 Qwen instead uses a separate three-epoch, rank-16 adapter with constrained category/location tokens. Adaptive tiling reuses this adapter. Qwen-Ground uses the visual-only initialization and two-stage optimization in Section IV; its backbone is not identical to direct Qwen’s.

B. Oracle Representations: How Much Does Context Matter?

On the same 4,778 positives, Qwen reaches 72.24% macro-F1 with a GT crop and 71.98% with ROI plus 20% context. A drawn ROI reaches 70.11%; masked background and coordinate-only conditioning fall to 68.64% and 67.34%. Coordinates alone therefore lose 4.90 points relative to the crop, while removing all surroundings can also discard useful event context. These fixed-adaptation comparisons isolate input representation, not localization performance or training-seed uncertainty.

C. Full-Frame Discovery: Ranking Is Not an Operating Point

D-FINE has the lowest N-FPR (2.14%) but recalls only 0.92% of positive frames at its saved cutoff. Grounding DINO reaches 90.14% P-AP yet only 0.11% contextual AP50. YOLO-World’s legacy zero-threshold run accepts every frame,

TABLE III

SESSION-DISJOINT ORACLE-REPRESENTATION RESULTS ON THE SAME 4,778 POSITIVE RECORDS (SEED 43). THE FIVE QWEN ROWS SHARE ONE QWEN3-VL-8B LoRA ($r = 16$) AND HIDDEN-STATE CLASSIFIER; WEIGHTS ARE FIXED ACROSS VIEWS. CTX. IS A 20% MARGIN. ALL ENTRIES ARE PERCENTAGES.

Model	Positive input	Acc.	Macro-F1	Valid
ConvNeXt [21]	GT crop (T3)	87.00	56.88	100
DINOv2 [18]	GT crop (T3)	90.08	66.39	100
Florence-2 [16]	GT crop (T3)	91.00	68.55	100
Qwen3-VL [17]	GT crop (T3)	91.38	72.24	100
Qwen3-VL [17]	drawn ROI (T2)	90.90	70.11	100
Qwen3-VL [17]	coordinates (T2)	90.10	67.34	100
Qwen3-VL [17]	masked bg. (T2 variant)	90.27	68.64	100
Qwen3-VL [17]	ROI + ctx.	91.13	71.98	100

producing 100% recall and 100% N-FPR; it is not a successful false-alarm-constrained operating point. Its AP50 remains below 0.01%. Presence ranking, rejection, and contextual localization are thus distinct failure modes, and low N-FPR cannot be read in isolation.

Direct Qwen leads the completed joint-discovery rows in P-AP (99.41%), AP50 (5.14%), C-F1 (44.80%), and G-mAP50 (6.10%). However, its validation-frozen cutoff produces 261 false alarms among 560 negatives (46.61%), with 99.79% positive recall. Qwen-Ground gives 3.57% G-mAP50 and 89 false alarms (15.89%), with 79.87% recall. Its median positive IoU is 0.236 and Recall@IoU0.5 is 23.19%, so contextual localization remains a bottleneck rather than a solved consequence of strong presence ranking. The D-FINE-DINOv2 cascade similarly reaches 23.70% candidate C-F1 but only 0.89% G-mAP50.

D. Adaptive Search and Calibration Transfer

Adaptive Qwen first inspects the global image. A global presence score below 0.85 triggers overlapping 2×2 tiles (10% overlap). The highest-presence candidate supplies a source-pixel crop with 20% context for reinspection. The final box remains the selected candidate’s box; the class comes from reinspection and the final presence score is the product of proposal and reinspection scores. This rule averages 2.63 visual calls per image, with a maximum of six.

Its test operating point gives 21 false alarms (3.75% N-FPR), but retains only 57.68% positive recall. G-mAP50 falls to 5.52% and C-F1 to 36.54%. Additional views therefore do not uniformly improve discovery; lower false alarms are accompanied by more missed positive frames.

The frozen direct-Qwen cutoff is 0.051623 and the tiling cutoff is 0.996302. Both were selected on validation to maximize recall subject to N-FPR $\leq 10\%$, attaining 9.01% and 9.88% validation N-FPR, respectively. On test these become 46.61% and 3.75%. A validation false-alarm constraint is therefore not a test-time guarantee. These observations concern the fixed acquisition-group split; they do not establish a causal explanation for the score shift or geographic negative transfer.

Recorded costs cannot support a speed ranking: direct Qwen uses batch-4 preprocessing/generation costs, Qwen-Ground uses batch-8 transfer/forward costs, and tiling sums cached-global and additional-view costs. Data loading, routing, and postprocessing are not consistently included. Consequently, Table IV prioritizes measured recall over heterogeneous latency; matched batch-one end-to-end timing is still required.

E. Matched Controls and Remaining Transfer Evaluation

Table V reports the full control and five completed one-change controls. The six registered controls remove heatmap guidance, multi-scale pooling, local ROI evidence, the classifier’s global-query feature, or the GT-to-predicted-box curriculum, or replace definition prototypes by a linear classifier. A new independent full control and all variants use 12 MS epochs, 12 CLS epochs, and a 10^{-4} head learning rate; geometric controls apply to both training stages. Archived incomplete-MS and eight-epoch runs are excluded. The full model reaches 37.79% C-F1, above all five completed controls. These recognition differences do not imply uniform joint-discovery gains: linear and no-ROI controls reach 3.66% and 3.62% G-mAP50 versus 3.57% for full. Without heatmap guidance, AP50 rises from 4.17% to 4.71%, while positive recall falls from 79.87% to 64.44% and N-FPR from 15.89% to 7.14%.

Removing the classifier’s global-query feature lowers both C-F1 (36.24%) and G-mAP50 (3.13%), while increasing N-FPR to 17.32%. Removing the box curriculum instead raises G-mAP50 to 3.95%, with 37.43% C-F1, 16.79% N-FPR, and 77.67% positive recall. The curriculum therefore has no consistent benefit across the measured objectives; it remains a disclosed training choice, not an established source of joint-discovery improvement. The registered full reference is retained rather than selecting an architecture on test. These single-seed observations do not establish statistical significance or module rankings. *Result placeholder: Complete the single-scale control.*

Table VI gives the measured session-disjoint reference for a fixed unseen-site and forward-temporal shortlist. Positive-only G^+ removes negatives before ranking; it is not the all-record G-mAP50. *Result placeholder: Insert measured cross-site/time results; the current paper does not claim demonstrated transfer.* The shared negative partition lacks aligned site/time semantics, so its false alarms remain a separate group-disjoint evaluation. Broader robustness, evidence-intervention, and scaling sweeps are future extensions, not prerequisites silently delegated to an external supplement.

VI. BENCHMARK DESIGN AND IMPLICATIONS

Context is not an object boundary. Each record asks whether an operational event is visible and which region supports it. The 0/1 ROI formulation penalizes unconditional event output without claiming exhaustive instance discovery. Strong oracle recognition but weak full-frame AP motivates search and localization diagnostics; it does not prove that a particular module is causally responsible. Likewise, IoU

TABLE IV

SESSION-DISJOINT RESULTS ON 4,778 POSITIVE AND 560 NO-EVENT TEST RECORDS (SEED 43; ALL SCORES IN PERCENT). N-FPR AND P-R ARE FALSE-POSITIVE RATE ON NEGATIVES AND PRESENCE RECALL ON POSITIVES AT THE SAME SAVED CUTOFF. AP50 AND G-mAP50 USE UNTHRESHOLDED, SINGLE-CANDIDATE PRESENCE-SCORE RANKING; C-F1 SCORES CANDIDATE CLASSES ON POSITIVES BEFORE PRESENCE GATING. LOW N-FPR ALONE DOES NOT ESTABLISH EFFECTIVE DISCOVERY.

System	Family/mode	P-AP $\uparrow$	N-FPR $\downarrow$	P-R $\uparrow$	AP50 $\uparrow$	C-F1 $\uparrow$	G-mAP50 $\uparrow$
D-FINE [15]	closed-set T1	84.86	2.14	0.92	2.29	–	–
Grounding DINO [13]	open-vocab. T1	90.14	5.18	5.94	0.11	–	–
YOLO-World [14]	open-vocab. T1 ^a	89.20	100.00	100.00	< 0.01	–	–
Qwen3-VL [17]	generative T4	99.41	46.61	99.79	5.14	44.80	6.10
Florence-2 [16]	generative T4	88.33	6.07	7.91	0.19	4.40	0.04
D-FINE + DINOv2 [15], [18]	modular T5	84.86	2.14	0.92	2.29	23.70	0.89
Qwen adaptive tiling [17]	global/tiles/crop	98.57	3.75	57.68	4.76	36.54	5.52
Qwen-Ground (ours)	feature-space T4	98.85	15.89	79.87	4.17	37.79	3.57

^aThe saved YOLO-World run uses threshold zero, not a validated 10%-N-FPR operating point. Qwen-Ground uses the completed 12-MS/12-CLS matched full control from Table V. Timing is omitted because recorded scopes and batch sizes differ. Bold marks the best nontrivial reported score, not a claim of dominance across the paired N-FPR/P-R operating points.

TABLE V

MATCHED QWEN-GROUND ABLATIONS (SESSION-DISJOINT, SEED 43; PERCENT). THE INDEPENDENT FULL CONTROL USES 12 MS AND 12 CLS EPOCHS; GEOMETRIC CONTROLS CHANGE BOTH STAGES, WHILE OTHER CONTROLS SHARE FULL-MS INITIALIZATION. RED ENTRIES ARE UNFINISHED. ALL ROWS USE THE SAME METRIC DEFINITIONS AS TABLE IV.

Variant	AP50	C-F1	G-mAP50	N-FPR	P-R
Full matched control	4.17	37.79	3.57	15.89	79.87
Without heatmap guidance	4.71	34.82	3.43	7.14	64.44
Single-scale features	TBD	TBD	TBD	TBD	TBD
Linear instead of prototype classifier	4.25	36.46	3.66	15.54	78.38
Without ROI features	4.34	35.61	3.62	14.29	75.16
Without global features	4.04	36.24	3.13	17.32	79.80
Without box curriculum	4.19	37.43	3.95	16.79	77.67

TABLE VI

CROSS-DOMAIN EVALUATION, SEED 43 (PERCENT). EACH MODEL IS TRAINED ON ITS PROTOCOL’S OWN TRAIN SPLIT. G⁺ RECOMPUTES G-MAP50 AFTER REMOVING NO-EVENT RECORDS FROM AP RANKING; IT IS NOT TABLE IV’S ALL-RECORD G-MAP50. NO-EVENT OPERATING POINTS ARE EVALUATED SEPARATELY; RED ENTRIES ARE UNFINISHED.

System	Session-disjoint		Unseen site		Forward time	
	C-F1	G ⁺	C-F1	G ⁺	C-F1	G ⁺
Direct Qwen3-VL	44.80	6.14	TBD	TBD	TBD	TBD
Qwen adaptive tiling	36.54	5.56	TBD	TBD	TBD	TBD
Qwen-Ground (matched initialization)	37.79	3.66	TBD	TBD	TBD	TBD

measures agreement with an accepted context region, not the causal sufficiency of its pixels. Evidence-removal interventions remain future work.

Report the whole decision, not one favorable number.

The main table separates presence ranking, paired false-alarm/recall operating points, candidate recognition, and event-aware grounding. Neither the low false-alarm detector nor the high-recall generative model dominates all of them. Qwen-Ground is a visual-only reference architecture, not the best measured end-to-end system. The five completed matched controls favor full-model candidate recognition, but removing the box curriculum improves event-aware grounding; removing global evidence harms both. The curriculum is thus an implementation choice, not a demonstrated joint-discovery

contribution. Only the single-scale control remains unfinished, and small single-seed differences do not establish module rankings. Different backbones and timing scopes preclude a controlled efficiency claim.

From frames to operational use. Mission-driven acquisition explains correlated images, grouped splits, and context-sensitive labels. T4/T5 evaluate the perception front end of the alert–review–report loop, not dispatch, remediation, or case closure. Because multiple frames can describe one event, image-level N-FPR is not operator workload. Deployment would additionally require temporal alert aggregation and measured review costs; the present results support decision assistance, not autonomous enforcement.

VII. LIMITATIONS, ETHICS, AND INTENDED USE

Each positive record supplies one primary contextual ROI; the benchmark does not measure exhaustive instances or segmentation. Diffuse ROI edges limit the interpretation of strict IoU; boundary-sensitivity and evidence-removal analyses remain unreported. The ontology is region-informed, rare classes remain difficult, unseen sites do not imply global coverage, and images do not evaluate tracking, causal explanation, or autonomous action. Experiments use one seed and session-disjoint testing; single-scale and cross-site/time evaluation remain pending. Small differences do not establish statistical significance or justify test-selected architectural changes.

The frame benchmark does not reproduce temporal alert clustering, live replay, dispatch, or report closure, so N-FPR is not itself operator workload. Proprietary interfaces, accounts, routes, and coordinates are not released; Figure 2 is a sanitized workflow abstraction.

The data support perception research and decision support, not autonomous enforcement. Authorization, privacy/figure review, and test-label sealing are complete. Public artifacts use opaque site/session IDs and exclude precise routes, coordinates, and unnecessary identifiers; protected test annotations remain private. Models may encode geographic or socioeconomic bias, so results must be read by site, time, class, and false-alarm slice rather than as a universal safety claim.

VIII. CONCLUSION

UTS-UAV links no-event rejection, contextual grounding, and recognition across five tasks with grouped deployment protocols. In session-disjoint evaluation, ROI plus context nearly matches GT-crop Qwen macro-F1 (71.98 versus 72.24), whereas coordinates alone lose 4.90 points. On the frozen session-disjoint test, direct Qwen reaches 6.10% G-mAP50 with 46.61% N-FPR; Qwen-Ground reaches 3.57% with 15.89% N-FPR. Adaptive tiling reaches 5.52% G-mAP50 and 3.75% N-FPR but retains only 57.68% positive recall. These results reveal a substantial gap between oracle recognition and reliable event discovery across operating points. The full model leads candidate recognition among five completed matched controls, but removing the box curriculum raises G-mAP50 to 3.95%. Global evidence benefits both measured recognition and grounding in this single-seed comparison; the curriculum does not offer a consistent gain. Cross-site and forward-time generalization remain unmeasured.

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